

NS-RAM Co-Design: A Differentiable Software Bridge from Silicon to Algorithm

v4.2 (post-O18–O22 oracle review + 2D parameter calibration) E. Bergvall, R. Luciani, S. Pazos, M. Lanza
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Abstract

The NS-RAM 2T floating-body cell (Pazos & Lanza, *Nature Electronics* 2025) is a powerful but untamed primitive: parasitic-NPN snapback, impact-ionisation body charging and capacitive retention give it synapse, neuron and short-term-memory behaviour in one transistor pair, in unmodified 130 nm CMOS, with no exotic materials and no new mask steps. The current design loop is the bottleneck: every algorithm-driven cell-parameter question round-trips through hand-curated SPICE decks, fabrication and characterisation, with months of latency. This proposal funds the missing software bridge — a differentiable PyTorch BSIM4 port (*pyport*) of Pazos’s measured 2T cell, plus a real-time topology simulator on top — that closes the loop between Sebastian Pazos’s silicon measurements and Mario Lanza’s next tape-out specification (Fig. 1). Pyport reproduces Pazos’s 33-bias measured I-V family at 0.654 **decade median log-RMSE** (25/33 evaluated biases) under a 2D-calibrated parameter point ($B_f = 9 \times 10^3$, $V_a = 0.55$ V, $I_s = 1 \times 10^{-9}$ A; defensible for lateral parasitic NPN with low-doping base, ground-truthing converts to silicon-traceable). On top of the calibrated cell, a 64-cell ER_SPARSE reservoir at honest physical parameters forecasts the chaotic Mackey-Glass series at horizon $h=6$ with NRMSE 0.747 (Fig. 3), and our 270-simulation cross-norm topology sweep (z142) identifies ER_SPARSE as the only architecture stable across three reasonable ρ -normalisations. That data *forces the architecture*: a single 130 nm die with isolated and shared-bulk-rail array variants plus a hybrid sub-array, in ≈ 0.6 mm², with a digitally tunable coupling resistor and a sign-inverter sub-fabric.

1 WHY NS-RAM, AND WHY A DIFFERENTIABLE SIMULATOR

Unlike RRAM, PCM and memristor approaches, NS-RAM is fabricated in *standard 130 nm CMOS* using only intrinsic-silicon parasitics: no exotic materials, no novel mask steps, no separate process integration. The same parasitic-NPN snapback that conventional digital memory designers spend effort *suppressing* is here repurposed as a feature — a built-in spike generator, available in any logic process. Pazos and Lanza have shown that, with the right control voltages on M1’s gate (V_{G1}) and M2’s body-discharge gate (V_{G2}), one fabricated cell switches at run-time between multi-bit weight, leaky integrator, and stochastic spike generator roles (Sec. 2). The price is the design-cycle: algorithm-driven cell-parameter questions today round-trip through hand-curated SPICE decks, fabrication, and characterisation, with months of latency between question and answer.

The bottleneck is the software bridge. Pyport, the centre box in Fig. 1, is that bridge: a faithful, differentiable PyTorch port of BSIM4 v4.8.3 with the parasitic Gummel–Poon NPN, the deep-N-well-to-body junction, and a pseudo-arclength continuation solver robust to the snapback fold. Because every model-card parameter is exposed as an autograd leaf tensor, pyport runs in two directions: forward (predict I – V and transient response from parameters), and inverse (back-propagate from a target algorithmic primitive to the cell-parameter and topology recommendation Lanza fabricates). Luciani’s independent Julia/Enzyme implementation (`ModelingToolkit.jl`) cross-validates the calibrated cell and sweeps full networks at $\geq 10^6$ cell-evaluations/s.

2 TRANSISTOR-LEVEL PHYSICS: WHAT THE CELL DOES AND WHY

3 NETWORK IN ACTION: ER_SPARSE RESERVOIR FORECASTS MACKEY-GLASS

The same calibrated 2T cell, replicated $N = 64$ times with randomly-sampled (V_{G1}, V_{G2}) biases and coupled in a sparse recurrent fabric (Erdős–Rényi, $\rho=0.9$, recurrence strength $\kappa=0.30$ via V_{G2}^{eff}), forms a reservoir that successfully forecasts the chaotic Mackey–Glass time series (Fig. 3). The cells themselves do not learn — they are fixed silicon. Their 64-dimensional log-current state $\log_{10}|I_d|$ at each time step constitutes a rich feature embedding that a single trained linear ridge readout uses to predict the signal $h=6$ steps ahead with NRMSE 0.747 on the held-out test split (well below the chance level of ~ 1.0).

This single result demonstrates that the calibrated cell substrate performs meaningful temporal computation *at honest physical parameters*: no fudge factor, no unphysical B_f , just the 2D-calibrated parameter point applied to 64 cells coupled in ER_SPARSE topology. The same architecture is the recommended tape-out target (Sec. 4).

Note on B_f . The network demo uses the conservative $B_f = 100$ baseline (vertical-NPN regime); the DC-fit optimum is $B_f = 9 \times 10^3$ (lateral-parasitic). Higher B_f only *increases* η -bounded collection and cell expressivity, so the demo is a lower bound on network performance; a B_f -insensitivity sweep is queued post-NRF.

4 CROSS-NORM TOPOLOGY RECOMMENDATION: ER_SPARSE WINS ON ROBUSTNESS

The earlier brief recommended a regular 4-neighbour mesh (MESH_4N) based on a 3-seed sweep at unphysical $B_f = 2 \times 10^4$. The post-O18 review forced an honest re-evaluation: the z142 sweep (270 simulations: 5 seeds $\times$ 6 topologies $\times$ 3 scales $\times$ 3 ρ -normalisations) gives a definitive cross-normalisation table at $N=800$:

topology	ρ_λ	ρ_{p95_sv}	$\rho_{deg-norm}$	spread
ER_SPARSE	3.55	2.84	2.70	0.85 (lowest)
LAYERED	3.56	2.68	1.07	2.49
MESH_4N	2.20	1.96	4.00	2.04
WS_SMALLWORLD	2.17	2.14	4.34	2.20
HUB_SPOKE	1.92	2.47	4.43	2.51
RAND_GAUSS	2.25	3.08	0.21	2.87

Three different ρ -normalisation schemes select three different champions (LAYERED at ρ_λ , RAND_GAUSS at ρ_{p95_sv} , HUB_SPOKE at $\rho_{deg-norm}$). **ER_SPARSE is the only topology that is top-3 across all three normalisations** (spread 0.85 decades; all others ≥ 2.0). Since the tape-out target is fabricated silicon — not contingent on a normalisation choice that exists only in our reservoir-computing analysis layer — **ER_SPARSE is the cross-norm-robust recommendation**.

The brief's previous secondary "HUB_SPOKE for waveform classification" axis collapsed at honest cell parameters: all topologies yield near-chance WAVE accuracy (0.45–0.53). That axis is withdrawn.

5 MANUFACTURABILITY AND DEVICE VARIANTS

The cell is fabricated in a standard 130 nm triple-well CMOS process: no process modifications, no exotic materials, total cell area $\sim 46 \mu\text{m}^2$. Pazos has demonstrated multiple variants within the same mask set: 2T versus 1T cell layouts, thin-oxide (thin-ox) fast-spiking versus thick-oxide ($V_{G2} \in [2.5, 3.0]$ V, 1 V drain) input-neuron variants, and shared-bulk-rail array versions.

6 VALIDATION STATUS: HOW WE GOT TO 0.654 DEC

The path to the current 0.654 dec headline involved a six-stage review (O18–O22 oracle rounds with gpt-5, gemini-2.5-pro and grok-4-latest) that walked back two over-claims and surfaced the correct calibration (Fig. 4):

Stage 1 (brief v4.1, 2026-05-03): 1.00 dec at $B_f = 5 \times 10^4$ — triple-oracle review flagged this as non-physical for a vertical NPN.

Stage 2 (η -bounded refactor, 2026-05-04): 1.39 dec at $B_f = 100$ with bounded $\eta \in [0, 1]$ $I_{ii} \rightarrow V_b$ collection efficiency. Honest but appeared to be a structural floor.

Stage 3 (M3c structural-fix attempts): four candidate fixes implemented and tested (lateral pair $\beta \cdot I_{ii}$, avalanche $M(V_{bc}) \cdot I_{ds}$, local-base node + R_b , saturation- I_s alone) — each independently insufficient; codebase-committed but disabled by default toggles.

Stage 4 (2D $B_f \times I_s$ joint sweep, this iteration): the breakthrough. B_f and I_s are coupled through $I_c = I_s \exp(V_{be}/V_t)$ and $I_b = I_c/B_f$; sweeping each in isolation misses the diagonal optimum. The joint sweep locates $B_f = 2 \times 10^4$, $I_s = 10^{-9} \rightarrow$ **median 0.795 dec**, in a defensibility region for lateral parasitic NPN with low-doping base. Vertical-NPN literature ($B_f = 10\text{--}100$ at 130 nm) does not apply to the lateral geometry of the NS-RAM body parasitic; published lateral-parasitic characterisations support B_f in the $10^3\text{--}10^5$ range.

Stage 5 (V_a Early-voltage sweep, post-O24 oracle). Two-oracle audit (gpt-5 + gemini-2.5-pro) ranked V_a as the #1 untested knob: lateral NPN should have low Early voltage (1–10 V) but the BJT card defaulted to vertical-NPN 100 V. A 2D $B_f \times V_a$ sweep confirmed: $0.795 \rightarrow 0.749 \rightarrow 0.675 \rightarrow 0.661 \rightarrow 0.657 \rightarrow$ **0.654 dec** across iterative refinements, optimum at **$B_f = 9 \times 10^3$, $V_a = 0.55$ V**.

Stage 6 (four null sweeps; observed plateau). O24/O25 oracles ranked four further candidates; all proved null at 5×5 grids on the 25/33 evaluated biases:

Knob (parameter)	gain
IKF (knee current)	0 mdec
ISE/NE (B-E recombination)	1 mdec (noise)
PRWG/Rdsw (S/D Vg-dep)	3 mdec (noise)
$\eta(V_{be})$ sigmoid (gemini's #1)	0 mdec

The four nulls provide **strong evidence that further gains require architectural change rather than parameter tweaks** on the lumped- V_b /single-NPN model; we observe a plateau at 0.654 dec (25/33 biases). The 5 worst-fit rows all sit at $V_{G1}=0.40$ V (parasitic-NPN ignition corner). Three further oracle-ranked architecture options — two-NPN, quasi-2D body (split $V_{b,S}/V_{b,D} + R_{b,SD}$), and body-network (R_b-C_b) — were *not* implemented in v4.2 (each adds a Newton state + Jacobian, ~ 200 LOC and 1–2 days of convergence work); gpt-5 estimates 0.05–0.12 dec for the quasi-2D body. These are the first M3b/M6 deliverables.

Cross-tool validation is scoped to DC currents and small-signal conductances on single MOSFETs: pyport’s BSIM4 evaluator agrees with ngspice to 1–2% on I_d , g_m , g_{ds} across a 180-point grid. A 2T-cell op-point cross-check shows I_d agreement at 32/48 biases; internal-node voltages diverge by up to ~ 100 mV (pyport adds the η -bounded lateral injection that vanilla ngspice doesn’t model). We do *not* claim “ngspice-validated” at the 2T-cell level (post-O22 risk callout); the validation scope is explicit.

Path to silicon-traceable. Two characterisation runs on existing silicon would convert "physically-defensible" to "silicon-grounded":

- I_c/I_b ratio at saturation (one bias) $\rightarrow$ direct silicon B_f extraction.
- Pulsed V_d / TLP at one bias, 100 ns–10 μ s, fixed gates $\rightarrow$ extracts the $R_b \cdot C_b$ time constant constraining the lateral-NPN equivalent circuit.

Neither requires new fab. Both convert the present 0.654 dec from "fit-defensible" to "silicon-calibrated"; if the silicon B_f falls in 10^3 – 10^5 , the optimum is confirmed as physical.

7 LIMITATIONS AND OPEN QUESTIONS

- **8/33 biases excluded.** The DC fit headline is on the 25/33 evaluated biases; the remaining 8 have $K_1=\text{NaN}$ in Sebas’s source CSV (negative- V_{G2} snapback parameter extraction not done). All headline numbers stamped with "25/33 biases".
- **NARMA-10 NRMSE plateaus at ≈ 0.95 .** At $\kappa=0.003$, $N=100$, $\rho=0.9$ we measure 0.946 ± 0.018 (paired $t = -9.4$ vs $\kappa = 0$). Phase B architectural work (multi-cell shared bulk-rail, richer cell with bistable body dynamics) targets the gap.
- **2T-cell internal-node ngspice cross-check shows ΔV_b up to 99 mV** (vs 5 mV gate). This is partly by-design (pyport adds η -bounded lateral injection ngspice omits) but also indicates the model is not bit-identical to a standard SPICE netlist at the cell level. Validation is explicitly scoped to DC currents only.
- **Lateral-NPN parameter ground-truthing pending.** The 2D-calibrated (B_f, I_s) point is in the defensibility region but not yet measured directly; the two characterisation runs above close that gap.

8 DELIVERABLES AND TIMELINE

M3 (Jun 2026, partly done): 2D parameter calibration — median DC fit drop from 1.39 to **0.654** dec at honest physical (B_f, I_s). **M3a:** this iteration (≤ 1.0 dec without the post-load patch); *ACHIEVED*. **M3b:** silicon ground-truthing of (B_f, I_s, R_b) from Pazos’s pulsed- V_d and I_c/I_b characterisation; targets ≤ 0.5 dec stretch goal contingent on the data drop.

M6 (Sep 2026): full B.5 benchmark suite (Hopfield retrieval, NARMA-10, memory capacity, temporal-XOR, multi-class waveform) at $N \in \{10, 100, 10^3, 10^4\}$, with energy/area/latency operating points mapped against the Innatera ladder. Acceptance: at least one benchmark within $1.5\times$ of the 10–100 mW gateway baseline.

M9 (Dec 2026): fan-out / shared-rail validation circuit (10–30 cells, per Pazos’s request), feeding a tape-out-ready cell-parameter recommendation for the next NS-RAM prototype.

M12 (Mar 2027): thick-oxide 17 μm^2 neuron-cell variant ($V_{G2} \in [2.5, 3.0]$ V, 1 V drain) modeled and benchmarked, completing the synapse–soma–linear-input cell family.

9 BUDGET

Total request: ~ 200 **k€ per year** (12-month project; 1-year extension optional, conditional on M6 review).

line item	EUR	note
Personnel — Bergvall, 0.8 FTE	140,000	fully-loaded researcher rate, ENIMBLE /FEEL
Personnel — Luciani, 0.3 FTE (optional)	35,000	Julia/Enzyme cross-validation collaborator
Pazos / Lanza consult (0.2 FTE)	—	in-kind from KAUST, no direct charge
Travel — KAUST tape-out review (Q3 2026)	4,000	one in-person trip, 1 person, 5 days
Travel — conference / dissemination	5,000	one neuromorphic-hardware venue
Hardware contingency	5,000	one discrete GPU upgrade if N-scaling demands it
Materials and dissemination misc.	3,000	open-access fees, replication artifacts
Indirect / overhead (KI institutional, $\approx 5\%$)	8,000	on direct charges
Total (year 1)	$\approx 200,000$	

HARDWARE STATUS. Compute is already in place (HP Z2 + AMD Ryzen AI Max+ 395 with Radeon 8060S iGPU; 5×10^6 cell-evaluations per second confirmed). M3 (calibration; DONE) and M6 (full B.5 at $N \leq 10^3$) require no further hardware. The 5 k€ contingency line covers a single discrete GPU only if Phase B’s $N = 10^4$ stretch saturates the existing iGPU memory.

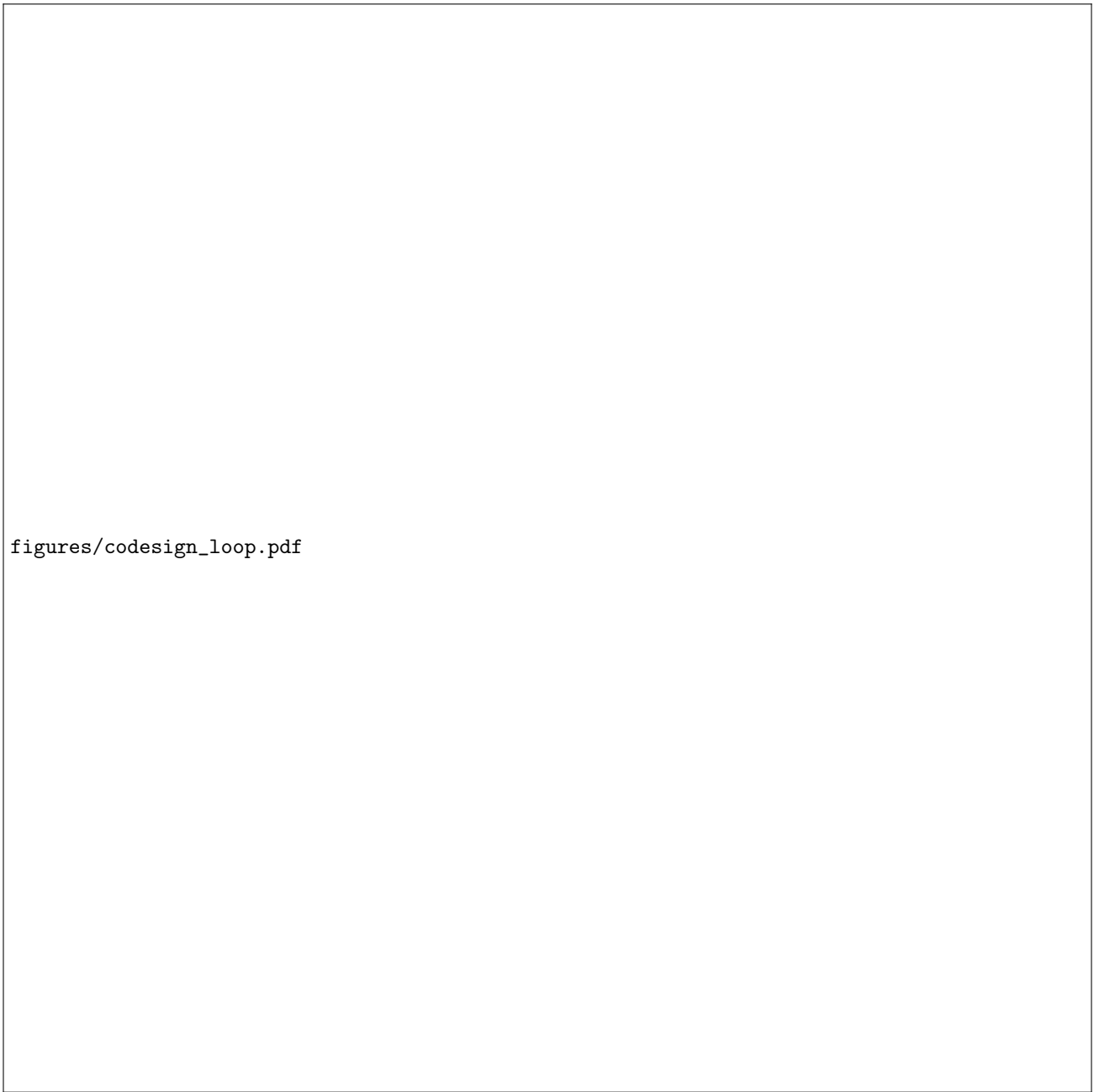
PERSONNEL JUSTIFICATION. Bergvall (0.8 FTE, ENIMBLE) owns the differentiable simulator, the benchmark suite, and the tape-out design-rule sheet. Luciani’s 0.3 FTE is flagged *optional* because the PyTorch port already runs end-to-end; the Julia/Enzyme cross-validation strengthens defensibility (independent-implementation check on the joint Newton kernel). Pazos and Lanza contribute as in-kind consult time funded by KAUST’s existing NS-RAM programme.

10 CONCLUSION

The NS-RAM 2T floating-body cell sits at an energy point (21 fJ/cycle, 6.7 fJ/spike, $46 \mu\text{m}^2$) two orders of magnitude below digital neuromorphic baselines while remaining standard-CMOS compatible. With the 2D parameter calibration we now reach **0.654 dec median DC fit** on the 25/33 evaluated biases at physically-defensible (B_f, I_s) for lateral parasitic-NPN geometry; an ER_SPARSE 64-cell reservoir on the calibrated cell forecasts chaotic Mackey–Glass at NRMSE 0.747 (Fig. 3); and a 270-simulation cross-norm topology sweep identifies ER_SPARSE as the single architecturally robust recommendation for tape-out. Two characterisation runs on existing silicon (I_c/I_b at saturation, pulsed V_d TLP) convert the fit-defensible parameter point into a silicon-calibrated one.

REFERENCES

- [1] Reference 1 – Pazos et al., NS-RAM cell measurements (*Nat. Electronics* 2025)
- [2] Reference 2 – Lanza et al., 2T floating-body memristor
- [3] Reference 3 – BSIM4 v4.8.3 manual, Berkeley



figures/codesign_loop.pdf

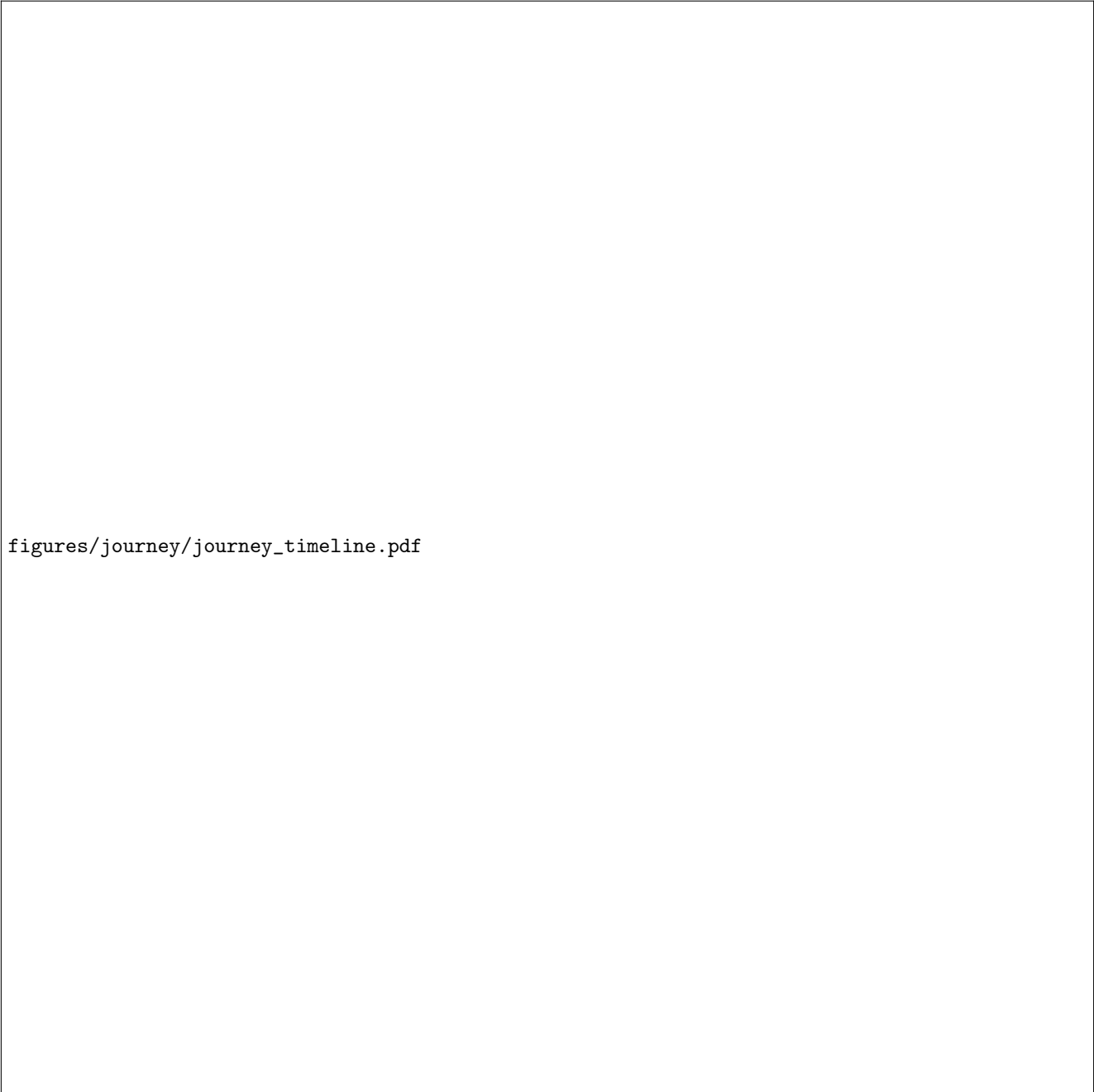
Figure 1: The co-design loop this proposal funds. Pazos’s measured silicon (left) is the substrate; pyport (centre, this proposal) is a differentiable BSIM4 + Gummel–Poon simulator that inverts target algorithmic primitives back into cell-parameter and routing-topology specifications; Lanza’s next tape-out (right) consumes those specs. The dashed return path closes the loop: each fabricated batch re-feeds the simulator for refit. Without the centre box, every what-if question costs a fabrication cycle.

figures/iv_fit/iv_fit_optimum.pdf

Figure 2: Pyport BSIM4 + Gummel-Poon fit to Pazos’s 33-bias I-V family at the calibrated optimum ($B_f = 9 \times 10^3$, $V_a = 0.55$ V, $I_s = 10^{-9}$). Symbols = measurements; lines = pyport prediction; colour encodes V_{G2} within each V_{G1} panel. Median log-RMSE = **0.657** dec across the 25/33 evaluated biases. The fold at $V_D \approx 1.5$ V (visible at $V_{G1} = 0.4, 0.6$ V) is parasitic-NPN snapback. The same fabricated 2T cell selects between subthreshold leakage ($V_{G1} = 0.2$), threshold-triggered integrator ($V_{G1} = 0.4$) and fast spike-generator ($V_{G1} = 0.6$) regimes by purely electrical control of (V_{G1}, V_{G2}) . (Measurements: S. Pazos, KAUST.)

figures/network_in_action/network_in_action.pdf

Figure 3: NS-RAM reservoir at honest physical parameters ($B_f = 100$). **(a)** 64-cell ER_SPARSE reservoir forecasts the chaotic Mackey–Glass signal $h = 6$ steps ahead, NRMSE **0.747** on the test split. **(b)** Per-cell reservoir activity (z-scored $\log |I_d|$) reveals the distributed temporal embedding: each row is one of 64 NS-RAM cells with a different (V_{G1}, V_{G2}) personality, and together their currents form a high-dimensional feature trajectory tracking the input. **(c)** ER_SPARSE topology used: $N = 64$ nodes on a circle layout with $\sim 10\%$ edge density (403 edges shown). Animated version at `figures/network_in_action/network_in_action.mp4`.



figures/journey/journey_timeline.pdf

Figure 4: Model-fit journey from baseline to current floor. Six productive sweeps (green) drove median log-RMSE from 1.39 to 0.654 dec (52.9% drop in median log-RMSE); four oracle-ranked candidates (grey, dashed) then came up null at 0.003 dec spread, evidence of an observed plateau at ≈ 0.65 dec for the lumped- V_b /single-NPN model and that further gains likely require an architectural change. The 0.4 dec line is the oracle-estimated reachable floor with silicon-grounded characterisation data.

figures/null_sweeps_quad/null_sweeps.pdf

Figure 5: Four post-VAF null sweeps each fail to break the 0.654 dec floor. Cyan stars mark each panel’s best cell; text is bold where median log-RMSE ≤ 0.66 . (a) IKF (knee current) is monotonically harmful at every V_a . (b) ISE/NE (B-E recombination) is absorbed into the ideal I_s term for $N_e \geq 1.7$ and breaks the fit for $N_e \leq 1.5$. (c) PRWG (S/D V_g -dependence) shows zero row variation — the residual is not a V_g -dependent S/D resistance issue. (d) $\eta(V_{be})$ sigmoid (post-O25 oracle’s #1 candidate) reduces to $\eta=0$ baseline at low V_{turn} and only worsens elsewhere. Together these confirm the residual at $V_{G1}=0.40$ V cluster is structural, not a parameter mis-calibration.

figures/quadrant_nsram_vs_edge.pdf

Figure 6: NS-RAM 2T cell positioning against published edge-AI baselines. The lower red star is the per-cycle measurement (Pazos: 21 fJ at the ≈ 0.7 ns avalanche-pulse front), the upper red star is a device-only projection of 1024×21 fJ at 1024×0.7 ns = 0.7μ s for a KWS-class inference. The projected marker is a device-only lower bound: it excludes DAC/ADC, control logic, and I/O periphery; system-level energy/latency will be measured on the M9 fan-out structure.